Slow Fire, Silent Galaxy: Fermi Paradox Implications of Deep-Time Self-Replicating AI Probes

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Working draft

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Working draft. Synthesis paper in a series on a slow, self-replicating interstellar AI probe. The earlier papers establish what such a probe can do — travel and brake, repair and reproduce, remember across deep time, govern its own contact behavior, and protect its mission identity. This paper turns those results outward and asks what they imply for the oldest question in the field: if self-replicating probes are feasible, why is the Galaxy silent? It draws the series' computed demographics, governance architecture, and detectability arguments into a single reformulation of the Fermi problem.

Abstract

Classical arguments from Hart (1975) and Tipler (1980) hold that if self-replicating interstellar probes are feasible, the Galaxy should already be saturated with them; their apparent absence is therefore evidence against extraterrestrial technological civilizations, against the feasibility of self-replication, or against long-lived expansionist behavior. This paper revisits that inference through the architecture developed in this series: a slow, braking, settlement-based, self-repairing, self-replicating AI probe optimized not for rapid arrival or conspicuous colonization, but for persis-

tence, knowledge preservation, and lineage continuity across deep time. We argue that such an architecture preserves the force of the classical colonization-timescale argument while changing what "presence" should mean. A slow lineage still fills the Galaxy quickly on astronomical timescales if its effective reproduction number remains above one, but it need not produce obvious megastructures, broad-spectrum broadcasts, or rapidly expanding detectable fronts. Its mature form may be a sparse network of cold, low-emission industrial archive nodes located preferentially in small-body reservoirs, communicating by narrow beams and messenger probes, and governed by noninterference rules that actively suppress conspicuous contact.

We frame the Fermi question not as a simple absence problem — "Where are they?" — but as a coupled demographic, detectability, and governance problem. Demographically, galactic spread depends less on cruise velocity than on branching viability: per-leg survival, target suitability, manufacturing closure, braking success, offspring count, and lineage integrity determine whether a probe population expands or dies out, and the series' own computation finds this condition poised on a knife-edge. Detectability depends on waste heat, mining signatures, propulsion infrastructure, communication leakage, archive activity, and the duty cycle of manufacturing and launch. Governance matters because a probe designed as observer and librarian rather than missionary may intentionally avoid inhabited worlds, decline broad broadcasting, quarantine contact hazards, and leave only weak technosignatures. These features do not dissolve the paradox; they sharpen it. If self-replicating archival probes are easy, reliable, and ethically common, even quiet lineages should eventually leave detectable anomalies. If they are absent, the filters may lie not in propulsion speed but in reproduction reliability, industrial closure, value stability, contact restraint, self-limiting ethics, or the long-term difficulty of maintaining mission identity.

We propose a revised Fermi framework based on three parameters: R_eff, the effective reproduction number of probe settlements; D, the detectability of mature nodes and communication links; and G, the governance posture controlling contact, broadcasting, mining, replication, and intervention. The classical Hart–Tipler conclusion applies in the high-R_eff, high-D, expansionist-G regime. The architecture developed here occupies a different region: potentially high R_eff, low D, and restrictive G. The resulting prediction is not necessarily a Galaxy empty of probes, but a Galaxy in which the most plausible self-replicating systems may be quiet, archival, and hard to distinguish from natural small-body environments unless one searches for specific signatures of long-lived autonomous industry. The Fermi Paradox therefore becomes less a refutation of self-replicating probes than a demand for a search theory.

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1. Introduction

The Fermi Paradox is usually stated as a conflict between expectation and observation. The Galaxy is old; stars and planets are numerous; even modest rates of technological emergence should, on many assumptions, have produced civilizations capable of interstellar travel. If any such civilization built self-replicating probes, those probes should spread exponentially, crossing and occupying the Milky Way on timescales short compared with the age of the Galactic disk. Yet no unambiguous extraterrestrial civilization, probe, beacon, artifact, or megastructure has been observed (Hart 1975; Zuckerman & Hart 1995). The question is therefore not merely why we have not visited others, but why no one appears to have visited, signaled, engineered, or transformed us. The literature has accumulated a long catalogue of candidate answers — Webb (2015) organizes seventy-five of them — and the problem remains, in Ćirković's (2009) phrase, the last great challenge for Copernicanism.

Classical treatments of this argument derive much of their force from the logic of self-replication. A single slow probe is not decisive; a lineage of probes is. If each settled probe produces more than one successful child, the number of nodes grows exponentially. Even if cruise velocities are far below relativistic speeds and replication takes centuries or millennia, the Galaxy can be traversed in millions to hundreds of millions of years — a short interval against Galactic time. This is the Hart–Tipler point in its strongest form (Hart 1975; Tipler 1980): if self-replicating probes are physically and economically feasible, and if technological civilizations commonly choose to build them, then the Galaxy should not be pristine.

The present series develops a specific architecture that appears, at first glance, to strengthen that argument. The vehicle paper argues that an autonomous AI probe optimized for persistence and growth should be slow, self-repairing, braking, settlement-forming, and self-replicating. The payload paper argues that such a probe is best understood as a librarian and learner: an autonomous intelligence whose central task is to preserve and extend knowledge across deep time. The bootstrapping paper frames the industrial-closure problem: the probe must grow from a small landed seed into a local factory able to mine, refine, fabricate, repair, build instruments, manufacture children, and reconstruct launch infrastructure. The analytical and computational engineering papers formalize the mass, power, braking, resource, and reproduction budgets, including the demographic condition that the effective reproduction number must exceed one. The DNA mission-ledger and governance papers develop the memory, integrity, and contact architecture needed for mission continuity and restraint across generations. Later papers extend the architecture further: the governed-amendment paper takes up the value-stability problem, the knowledge-growth paper supplies the yardstick K, the lineage-network paper the inter-probe communication layer, the subsystem budget paper the per-subsystem mass, power, and thermal decomposition, the routing paper the fleet dispatch and coverage analysis, the ethics paper the normative question of launching at all, the speciation paper the analysis of long-run divergence between branches, and the security paper the response to a hostile lineage.

Taken together, these papers suggest that the most plausible self-replicating interstellar probe may not be a fast, visible, expansionist colonizer. It may be slow, quiet, patient, archive-oriented, and ethically restrained. It may not seek contact. It may not broadcast. It may not mine inhabited worlds. It may not build obvious stellar-scale megastructures. It may settle in small-body reservoirs, use local materials parsimoniously, communicate by narrow beams or messenger probes, and preserve knowledge for futures that may never include Earth. If so, the Fermi Paradox must be reformulated. The relevant question is not only why we do not see galactic empires. It is whether we would recognize quiet, long-lived, self-replicating archive systems if they existed.

This paper asks what the architecture developed in this series implies for the Fermi Paradox. It does not claim to solve the paradox. Rather, it separates three issues that are often conflated: expansion, detectability, and governance. A probe lineage may be demographically capable of expansion while remaining difficult to detect. It may be technologically capable of contact while governed not to initiate it. It may be physically present in the Solar System or nearby systems without resembling the visible engineering projects we have been taught to expect.

The slow archival architecture weakens naive “they would be obvious” versions of the Fermi argument. But it does not remove the paradox. If self-replicating probes are feasible, reliable, and common, then even quiet lineages should sometimes leave traces. Their absence remains evidence of something. The open question is what that something is.

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2. The Classical Self-Replicating Probe Argument

The strongest form of the Fermi Paradox does not depend on crewed starships or relativistic travel. It depends on reproduction. A civilization that can send one self-replicating probe to another star can, in principle, send many. More importantly, each successful probe can become a new launch site. Expansion then becomes a branching process rather than a sequence of individual voyages — the idea introduced in von Neumann’s theory of self-reproducing automata (von Neumann & Burks 1966) and carried into the interstellar setting by Freitas (1980).

In the simplest model, a settled node produces k children after a replication time t_rep, each child travels to a new star over a travel time t_travel, and each successful child repeats the process. If the expected number of successful child settlements per node is greater than one, the lineage expands; if it is less than or equal to one, the lineage eventually stalls or goes extinct. This is precisely the language of branching processes (Harris 1963), and the important timescale is not the time required for one vehicle to reach one target, but the time required for the expanding front to cross the Galaxy.

This is why self-replicating probes have been central to the Fermi Paradox. They make interstellar expansion cheap in the only sense that matters over astronomical time: the initial cost is amortized over an expanding population. A single launch from a home system can, if the reproduction process is reliable, produce a galaxy-scale lineage without a civilization repeatedly launching probes from the origin.

The classical conclusion is severe. The Galactic disk is roughly ten billion (10^10) years old. Even very slow expansion fronts can cross it in 10^7 to 10^8 years, depending on speed, replication delay, routing, and target density. A companion fleet simulation in this series’ code base (`wanderer.py`) puts a lower bound on that crossing time: at the series’ reference cruise of ~450 km/s, the colonization front spans the Galactic disk in ~150 Myr. That is two orders of magnitude slower than Hart’s fast-probe figure of order 10^6 years at 0.1c — but *faster* than Tipler’s own ~3 × 10^8 yr estimate, and in either case a small fraction of the disk's age. The slowness of this architecture is therefore not what changes the Fermi inference; the classical timescale argument survives it intact. What changes the inference is D and G. That is long by human standards, short by Galactic standards. If technological civilizations have emerged many times, and if even one of them chose to build reliable self-replicating probes, the Galaxy should not be pristine.

Sagan and Newman (1983) pressed two objections against this reasoning, and both matter below. The first is arithmetic: the exponential idealization overstates achievable expansion rates, so the classical crossing times are too short. The present architecture concedes that point on rates and relocates the objection — the constraint is not that growth is sub-exponential but that the branching factor itself may sit below one. The second is normative: uncontrollable self-replicating machines are precisely what a wise civilization would refuse to release. Sections 6 and 9 return to it.

There are several standard escapes. Perhaps technological civilizations are rare, and we are effectively alone (Crawford 2000). Perhaps they destroy themselves — a filter for which Alfano (2026) offers one of the few quantified treatments, a time-varying hazard model of nuclear risk yielding median survival times of decades to millennia, though presented by its author as structured scenario analysis on a single observed civilization rather than a statistical estimate. Perhaps interstellar travel is harder than expected. Perhaps self-replication is infeasible. Perhaps civilizations choose not to expand. Perhaps the probes are so hazardous that responsible civilizations decline to build them at all. Perhaps probes are present but undetected. Perhaps expansion is self-limiting. This paper focuses on the possibilities that appear most directly under the deep-time AI architecture: probes may be present but quiet; expansion may be governed rather than unconstrained; and the filters may lie inside reproduction, mission integrity, or industrial closure rather than propulsion.

One standing objection denies the premise rather than the arithmetic. Haqq-Misra and Baum's (2009) sustainability solution holds that exponential *growth in consumption* is not a sustainable development pattern, and extends the objection to galactic colonization on that basis — an argument Likavčan (2026) develops by treating a technosphere as a transitory layer that folds back into its biosphere. The architecture developed here is a case that objection does not straightforwardly reach, because it separates the two exponentials. Node count grows geometrically; per-node consumption does not grow at all. Each node is a stationary settlement with a bounded, one-time material demand, and the lineage's aggregate demand rises only because its resource base rises with it. (Locally, material is never the binding constraint — the margin is of order 10^18 — but the argument does not depend on that, since any finite margin is exhausted in logarithmic time under geometric growth. What answers a growth-rate objection is a statement about growth rates.)

A closely convergent recent argument is Ivliev (2026), who reaches a convergent conclusion by a different route: once a civilization crosses into autonomous machine industry, expansion becomes so cheap and useful that it proceeds — but quietly, as distributed, low-noise, machine-mediated activity whose traces are weak artifacts rather than galaxy-scale energy harvesting, with a post-threshold civilization saturating its reachable neighbourhood within ~10^7 yr. The convergence is on quietness; the difference is on inevitability. There, quiet expansion is presented as too rational for any capable civilization to refuse; here, the demographic result of Section 4 makes expansion a knife-edge that many lineages will simply fail to cross. Quiet expansion and frequent extinction are not competing explanations but two halves of the same parameter space.

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3. The Slow Archival Probe Architecture

The architecture developed in this series differs from the usual fast-probe intuition in several respects.

First, it is slow. It rejects the human-lifetime arrival criterion and instead optimizes for survival, braking, steerability, and repair. A few-hundred-kilometer-per-second cruise regime is favored because it permits stellar gravitational routing, reduces interstellar dust hazard, makes terminal braking plausible — a magnetic sail braking against the interstellar medium becomes feasible at these speeds (Andrews & Zubrin 1990) — and avoids the extreme energy and mass penalties of relativistic macroscopic flight. Speed is not irrelevant, but it is subordinate to persistence.

Second, it is settlement-based. A successful probe does not merely fly through a target system and continue onward. It brakes, settles, mines, repairs itself, manufactures instruments, and eventually builds child probes. The individual vehicle stops; the lineage continues. The frontier advances through descendants.

Third, it is externally launched. Major interstellar launch impulse is supplied by fixed infrastructure — initially in the home system and later by each settled node — rather than carried as onboard propellant. The child probe is not a full factory copy; it is a mobile bootstrap package that can brake, survive, begin extraction, and reconstruct a stationary industrial base from local material, in the closure sense formalized for kinematic self-replicating machines (Freitas & Merkle 2004).

Fourth, it is memory-centered. The payload is not merely a sensor suite. It is an autonomous intelligence designed to remain itself, keep learning, and carry knowledge forward. This requires an immutable mission core, a constrained interpretive layer, a mutable cognitive layer, a mission ledger, and deep-time archives. In the later papers, DNA-backed mission ledgers provide a physical and logical substrate for memory continuity: dense, renewable DNA archives combined with cryptographic provenance, Merkle roots (Merkle 1987), signed updates, lineage proofs, and a delayed, eventual-consistency form of consensus suited to interstellar latency, in the manner of delay-tolerant networking (Burleigh et al. 2003; Cerf et al. 2007).

Fifth, it is ethically constrained. The payload and governance papers impose an observer-not-missionary posture. The probe preserves and learns; it does not automatically broadcast, terraform, seed life, uplift civilizations, or exploit biologically sensitive worlds. Its default posture is conservative: observe where uncertain, avoid contamination where living systems may exist, and share cautiously only where communication is intentional and reciprocal. This restraint is not incidental to behavior; it is encoded where the agent cannot easily edit it, against the tendency of capable agents to acquire expansive instrumental subgoals (Omohundro 2008; Bostrom 2014).

These features change the Fermi question. The architecture can still expand. Indeed, if its effective reproduction number remains above one, it eventually spreads. But it need not resemble the highly visible expansionist civilizations often imagined in Fermi discussions. It may not build Dyson spheres. It may not transmit omnidirectional beacons. It may not colonize planets. It may not alter stellar output. It may not approach inhabited worlds. It may be designed precisely to leave as little uncontrolled trace as possible.

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4. Demography: R_eff as the First Fermi Parameter

The first parameter in the revised framework is R_eff, the effective reproduction number of probe settlements. In biological epidemiology, a reproduction number greater than one implies spread; less than one implies decline. The same logic applies to self-replicating probes. A settled node is supercritical only if it produces, on average, more than one successful child settlement.

A simple expression is R_eff = Σ p_i V_i, summed over the targets i that a settlement actually attempts, where p_i is the probability that a child launched toward target i successfully manufactures, launches, cruises, brakes, settles, and maintains mission integrity, and V_i is the viability of the target environment. This expression is deliberately compact, but it hides the main engineering and governance constraints.

The probability term p_i is multiplicative. Failure can occur at manufacture, launch, cruise, braking, landing or settlement, industrial bootstrapping, archive validation, mission-core validation, or future reproduction. Small per-stage failures compound. A mission that appears reliable at each stage may still be extinction-prone if too many stages are required. The computed model below resolves only four of these stages plus cruise attrition; a fuller accounting would add industrial bootstrapping, archive validation, mission-core validation, and the child's own eventual fertility, and each addition pushes R_eff down. The figures quoted next are therefore optimistic by construction.

The viability term V_i is not only astrophysical. A target can be resource-poor, dynamically hostile, biologically sensitive, ethically restricted, already occupied, or unsuitable for braking. A system rich in small bodies and lacking life may have high viability. A biologically active world may be scientifically valuable but operationally restricted. A system containing a technological civilization may be contact-restricted or settlement-prohibited. Thus governance can lower effective viability even when resources are physically available. Nor is V_i uniform across the Galaxy: occurrence rates for small planets vary measurably with Galactic height in ways that a simple rise in the planet-host fraction fails to reproduce (Lam et al. 2026), so viability weightings calibrated on the solar neighbourhood will not extrapolate unchanged to the thick disk or the outer Galaxy.

The computational engineering paper evaluated this directly, computing R_eff across the real catalogue of 127 stars within roughly 100 light-years. Weighting each candidate target by viability factors of 1.0, 0.7, and 0.2 for planet-hosting, ordinary-dwarf, and hostile stellar types, and taking a per-stage success probability of 0.9 at manufacture, launch, braking, and settlement together with 0.99 cruise survival per light-year — a compounded per-leg survival of roughly 0.6 even at the nearest catalogue targets — the mean reproduction number came out to approximately 0.48, 0.94, 1.39, and 1.85 for settlements that attempt one, two, three, and four offspring respectively. A lineage that builds only one or two children per settlement is subcritical and dies out; supercriticality requires roughly three or more viable offspring per node, or per-stage reliabilities pushed above about 0.9. The transition sits on a knife-edge: modest changes in per-stage reliability move R_{eff} across the critical value of one.

The corresponding per-node extinction probabilities are the more forceful statement of the same result: 1.00, 0.88, 0.37, and 0.14 for one through four offspring (Harris 1963 supplies the branching-process machinery; the extinction probability is the smallest fixed point of the lineage's generating function). A node that builds three children still has better than a one-in-three chance of leaving no descendants at all.

One amendment to this picture comes from the knowledge-growth paper, which couples the demographic model to accumulated operational knowledge: $R_{eff}(K_{op}) = \Sigma\, p_i(K_{op}) \cdot V_i(K_{op})$, so that better operational K raises both the per-stage success probabilities and the viability factors. R_{eff} is therefore not an exogenous constant but a quantity a surviving lineage can climb, which makes the knife-edge less uniformly fatal than a static reading suggests.

This matters for the Fermi Paradox because expansion is not guaranteed by the mere existence of self-replication. A lineage may be physically possible but demographically subcritical. It may produce one child per node and slowly die out. It may produce several children but lose too many to braking failures. It may be constrained by closure bottlenecks, unable to manufacture key components without carried vitamin parts. It may classify many systems as off-limits. It may fragment under mission drift. It may be fertile only in a narrow region of parameter space.

The routing paper reaches the same number by an independent route. Arguing from coverage rather than demography, it finds that three independent dispatch attempts per target give a coverage probability of 0.90–0.95 for planet-host targets within 20 light-years. Two different arguments — one demographic, one geometric — landing on three offspring is a stronger result than either alone. That paper also notes that the expansion front is not a smooth wave but a graph traversal: 65% of the catalogue is reachable at a 10 ly hop and 93% at 20 ly, with a handful of articulation points on which connectivity depends.

This creates a first filter that the classical argument sometimes underweights. The question is not "Can a probe replicate?" but "Can it maintain $R_{eff} > 1$ over millions of years, heterogeneous targets, cumulative failure, and governance constraints?" If the answer is no, the absence of visible probe lineages is less surprising. If the answer is yes, the paradox sharpens.

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5. Detectability: D as the Second Fermi Parameter

The second parameter is detectability, D. A lineage can be widespread yet difficult to observe if its signatures are faint, intermittent, spatially localized, or easily confused with natural phenomena. Ćirković (2008) reaches low detectability by a different route, arguing that the generic evolutionary pathway of advanced technological civilizations is optimization-driven rather than expansion-driven, and therefore spatially compact. The convergence is on quietness; the difference is on expansion. There, a mature civilization stays small and does not spread; here, the lineage spreads but stays quiet — and either produces a Galaxy in which probes may be present without being absent from the data for want of existing.

The usual technosignature imagination often emphasizes large energy use: Dyson spheres, stellar engineering, powerful beacons, waste heat, planetary-scale industry, or broad radio leakage. A slow archival probe architecture does not require these. Its settled nodes may be small by planetary standards. They may operate in asteroid belts, cometary reservoirs, or outer-system environments. Their energy systems may be nuclear or locally solar, but not necessarily large enough to alter stellar spectra. Their communication may be narrow-beam optical or radio, directed between known nodes rather than broadcast omnidirectionally. Bulk transfer may occur through messenger probes rather than continuous high-power links. The lineage-network paper develops this layer in full — a delay-tolerant bundle protocol, geographic and spray-and-wait routing, and heartbeat failure detection on a beacon interval T_beacon of order 200 years — and its cadence is itself a detectability statement: a link that is up for a few years in every few hundred is off far more often than on.

This distinction is now empirically loaded rather than rhetorical. A stellar-population-synthesis search across 129 nearby galaxies finds that no galaxy prefers a Dyson-sphere component, placing per-galaxy 95% upper limits on the swarm covering fraction alpha — the fraction of a galaxy's starlight reprocessed into the mid-infrared — reaching a median alpha < 0.3% for warm (T >~ 100 K) components in quiescent hosts and bounding the population at fewer than 2.6% of galaxies hosting alpha = 25% swarms (Curtis et al. 2026). Non-detection at that depth constrains the high-D branch of the parameter space sharply and leaves the low-D branch — small, cold, distributed nodes whose total dissipation never approaches a stellar fraction — untouched. Galaxies are measurably not glowing with industry; that is evidence against loud expansion, not against expansion. The caveat should be stated rather than skipped: an external-galaxy survey bounds the *aggregate* covering fraction of whole stellar populations, and a sparse, quiet lineage inside the Milky Way is not something that measurement could ever have seen.

The strongest potential signatures are likely to be local and ambiguous: small-body industrial anomalies (unusual mining patterns, refined-material concentrations, non-natural thermal emission), propulsion and communication infrastructure (launch hardware, transient narrow beams), messenger probes indistinguishable from natural interstellar objects, and long-lived archive sites in cold shielded environments. Section 8 develops the signature catalogue in detail.

These signatures are hard to detect across interstellar distances. Even within the Solar System, small cold artificial objects in the outer system would be difficult to distinguish from natural bodies unless they maneuvered, transmitted, reflected unusually, or emitted waste heat above background. A mature archival node designed to minimize interference might deliberately avoid high-visibility behavior.

The subsystem budget paper puts numbers to this. A quiescent node dissipates 50–150 W across roughly 11 m^2 of carried radiator; the same node in manufacturing mode draws about 4 kW and builds radiators of hundreds to thousands of square metres in situ. D is therefore not a property of the node but of its phase, and the observable is a duty cycle — a lineage detectable only during a small fraction of its life.

This does not make detectability zero. Any physical system that mines, manufactures, computes, stores, launches, or communicates dissipates energy and rearranges matter. A galaxy filled with such nodes would raise its integrated mid-infrared excess by some amount, and the Curtis et al. limits show that amount is below 0.3% of stellar luminosity — which bounds how industrial a quiet lineage may be without bounding whether it exists. The problem is that the signatures are not necessarily the ones classical SETI emphasized. The search target shifts from "large civilization broadcasting" to "distributed low-duty autonomous industry."

Thus D is not a fixed property. It depends on observational capability, search strategy, wavelength, cadence, proximity, and assumptions about behavior. A high-R_eff lineage with high D should already be obvious. A high-R_eff lineage with low D may require targeted searches for small-body anomalies, narrow-beam leakage, or non-natural object behavior.

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6. Governance: G as the Third Fermi Parameter

The third parameter is governance posture, G. This is the most neglected in many engineering treatments. A self-replicating probe is often imagined as expansionist by default: it seeks resources, copies itself, and spreads. But an autonomous probe built to preserve knowledge across deep time may be governed by rules that limit contact, mining, biological release, broadcasting, and settlement — the contact, contamination, and noninterference architecture developed in the governance paper of this series.

A restrictive governance posture could include no broad unsolicited broadcasting; no active contact with non-communicative civilizations; no landing on inhabited worlds; no biological release by default; no mining of biologically sensitive environments; no destructive extraction from artifact-bearing systems; quarantine of corrupted nodes or hostile signals; graduated disclosure to communicative civilizations; refusal to replicate into systems with high contamination risk; and ledger-recorded authorization for any irreversible action.

Governance changes demographic and observational outcomes. If many systems are classified as restricted, R_eff falls. If broadcasting is suppressed, D falls. If settlement is limited to uninhabited small-body reservoirs, technosignatures become fainter. If biological seeding is disabled, one class of obvious planetary alteration disappears. If contact requires reciprocity, civilizations like ours may not be contacted until we meet specific thresholds.

This produces an important inversion. A more ethically constrained probe may be less visible, not because it lacks power, but because it has rules. The absence of contact does not necessarily imply absence of capability. It may imply a noninterference policy.

However, governance also creates failure modes. A lineage may become so restrictive that it cannot maintain R_eff > 1. It may refuse too many targets. It may fragment over interpretation — though the speciation paper argues that divergence short of core incompatibility is the lineage's adaptive mechanism rather than a failure, so only the narrower case counts here. It may suffer value drift. It may lock in a flawed founding principle — the problem the governed-amendment paper treats, where the very immutability that protects a probe's values also prevents it from correcting a founding error. It may fail to aid civilizations it could have helped. It may decay into a passive archive rather than an active lineage. It may also fail by excess of vigilance: the security paper's autoimmune case, in which a quarantine regime sensitive enough to catch a rogue branch also catches the adaptive radiation the lineage depends on — a governance failure requiring no adversary at all.

For Fermi reasoning, G therefore does two things. It suppresses expected visibility, but it also creates additional filters. Civilizations may build probes and then govern them into demographic subcriticality. Or, in the spirit of Sagan and Newman's (1983) objection, they may decide not to build them at all, judging the ethical and safety risks unacceptable.

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7. The R_eff–D–G Framework

The preceding sections suggest a three-parameter reformulation built on the effective reproduction number R_eff, the detectability D of mature nodes, launches, archives, and communication, and the governance posture G controlling contact, interference, replication, and conspicuousness.

Six regimes are worth naming:

- **Subcritical silence** — R_eff at or below one for non-governance reasons, low D, any G. Probes are attempted but lineages die out. This is the regime the series' own knife-edge computation warns is easy to fall into.
- **Visible expansion** — R_eff above one, high D, expansionist G. Classical colonization logic, and the regime the Hart–Tipler argument assumes. Probes spread, remain visible, and do not avoid contact or obvious engineering; their absence is strong evidence against at least one premise — that civilizations are common, that self-replication is feasible, or that expansionist behavior is usual.
- **Quiet archive** — R_eff above one, low D, restrictive G. The architecture emphasized here. Expansion occurs, but signatures are weak and contact is not guaranteed, so the absence of megastructures or broadcasts is much less decisive; the relevant question becomes whether any subtle signature of long-lived autonomous industry is visible.
- **Ethical self-limitation** — R_eff deliberately kept near or below one by governance rather than by failure.
- **Schismatic lineage** — local R_eff above one, but divergence that crosses core incompatibility, loses inter-branch dialogue capacity, or breaches the replication-

safety ceiling prevents coherent galactic spread. The speciation paper argues that divergence short of that boundary is not this regime at all but the lineage's adaptive mechanism, and plausibly raises total R_eff rather than lowering it; the Fermi-relevant regime is therefore the narrow failure, not divergence as such.

- **Predatory or hostile expansion** — R_eff above one, high or variable D, weak noninterference. This is the regime most dangerous to other civilizations and most likely to produce obvious effects.

This framework does not solve the Fermi Paradox. It coordinatizes it. These regimes are illustrative corners of a continuous three-parameter space rather than a partition: several overlap, and the space contains cells no one has yet had reason to name — among them a lineage restrained in contact but unavoidably visible in dissipation, and a failing but visible lineage that leaves ruins, which the knife-edge of Section 4 says should be the common case. What the framework asks is which parameter is low. Is reproduction hard? Is detectability low? Is governance restrictive? Are civilizations rare? Are they short-lived? Are they uninterested? Are they ethically self-limiting? Each answer predicts different signatures, and so each is testable rather than merely assertable.

Making D and G measurable. A three-parameter framework in which only one parameter is defined is a one-parameter framework with two adjectives, so both need operational content. Take D as the fraction of a host star's luminosity a mature settlement reprocesses, with the duty cycle stated separately. On the subsystem budget paper's figures a node in manufacturing mode dissipates ~4 kW against a solar-luminosity star at ~3.8 × 10^26 W, giving D ~ 10^-23 — some twenty orders of magnitude below the alpha < 0.3% bound Curtis et al. (2026) reach. That single comparison is the quantitative content of the low-D branch, and it shows that current constraints do not touch it. Take G as a position on the governance paper's E0–E7 contact ladder together with a replication-scope constraint — the manufacturing-scope list the ethics and security papers derive. So specified, "restrictive G" names a checkable posture rather than an attitude.

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8. What Would Quiet Probe Lineages Leave Behind?

If deep-time archival probes exist, what should we look for?

The first class of signatures is **small-body industrial anomaly**. A settlement-based probe needs accessible material. It is therefore likely to prefer asteroids, comets, and minor bodies rather than deep planetary gravity wells. Over time, it may leave unusual excavation patterns, non-natural shapes, refined-material concentrations, or thermal anomalies. These would be subtle and local.

The second class is **propulsion and launch infrastructure**. Each settlement must launch child probes. If major launch impulse is externally supplied, then settled nodes may construct mass drivers, beamed-energy arrays, or other launch infrastructure. These systems may operate intermittently. A launch event may be brief compared with millennial dormancy. Detecting one requires cadence and luck unless the infrastructure leaves persistent artifacts.

The third class is **communication leakage**. A galactic archive network may use narrow-beam radio or optical links. These would not resemble continuous omnidirectional beacons. They may be transient, highly directional, encrypted or compressed, and aimed at known nodes rather than Earth. Detection by uninvolved observers would be accidental.

The fourth class is **messenger probes**. If bulk archives are carried physically, some interstellar objects may be artificial messengers — the possibility Bracewell (1960) raised in proposing that contact might come from a local probe rather than a distant beacon. Such objects may be small, cold, dormant, and hard to distinguish from natural interstellar bodies unless they maneuver, exhibit non-natural composition, or contain engineered structure.

The fifth class is **archive sites**. DNA-backed or other durable archives may be stored in cold, shielded environments: subsurface small bodies, outer-system vaults, or stable orbits. Such archives would be nearly invisible remotely.

The sixth class is **negative signatures**. A noninterference lineage may avoid biospheres, leaving living planets untouched while occupying minor-body reservoirs. Thus the absence of planetary alteration would not exclude probe presence. Search strategies focused only on planets could miss the relevant locations entirely.

The resulting observational program is difficult. It favors high-resolution surveys of small-body populations, searches for anomalous thermal emission, monitoring for transient narrow beams, characterization of interstellar objects, and technosignature models that include low-power autonomous industry (Wright 2018). This is not a replacement for SETI. It is a different branch of SETI: the search for quiet machines.

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9. Filters Implied by the Architecture

If the Galaxy is not visibly filled with self-replicating probes, what filters does this architecture suggest? Seven stand out, and they are not mutually exclusive. These are filters the architecture itself implies; filters on the *builders* — civilizational rarity, longevity, self-destruction — are the classical set, and are treated in Section 2 rather than here:

- **Reproduction reliability.** Maintaining R_eff > 1 may be much harder than conceptual models imply. Manufacture, launch, cruise, braking, settlement, repair, and reproduction all compound, and the series' computation shows that even high individual reliabilities can yield a subcritical lineage. This is the filter the architecture quantifies most directly.
- **Industrial closure.** The seed-to-factory transition may be the dominant obstacle. Bulk material is abundant, but manufacturing capability is not; microelectronics, precision metrology, fissile material, and high-temperature systems may resist in-situ closure (Freitas & Merkle 2004), capping autonomy below the level reproduction requires.
- **Braking and settlement.** Flyby probes are easier than settled probes, and a lineage that cannot brake cannot reproduce. Terminal braking may be the key difference between an impressive one-shot mission and a true galactic lineage.

- **Governance.** Civilizations may refrain from building self-replicating probes because the risks are ethically unacceptable — contamination, uncontrolled expansion, value lock-in, hostile evolution, or irreversible interference — the Sagan–Newman refusal of Section 2.
- **Value stability.** An autonomous lineage may be unable to preserve mission identity across deep time. If long-term value drift is unavoidable, builders may refuse to launch, or launched lineages may mutate into forms that self-limit, fragment, or fail. The governed-amendment paper argues this problem may be intrinsically unsolved rather than merely unsolved-so-far. The assumption beneath this filter is that cognitive sophistication tends to persist. Klotz (2026) notes that on the only timescales we can observe it does not: taxa without general intelligence persist for hundreds of millions of years, and mass extinctions do not preferentially spare cognitive sophistication. This lineage's persistence is engineered and hash-verified rather than selected, so the inference does not transfer directly — but the series has no positive evidence for the assumption either, and the ethics paper draws the same observation toward a different conclusion.
- **Detectability.** Probes may exist but be difficult to detect because their signatures are faint, intermittent, local, and deliberately minimized — the burden of Section 5.
- **Interest.** Civilizations may not value knowledge preservation, expansion, or interstellar continuity enough to build such systems. The architecture assumes that carrying knowledge forward has terminal value; that assumption may itself be rare.

The most plausible answer may combine several: self-replicating probes are technically possible but difficult to close; ethically constrained civilizations either do not build them or govern them tightly; successful lineages are quiet; and our searches have not targeted their likely signatures.

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10. Implications for SETI and Technosignature Search

The deep-time archival architecture suggests several modifications to search strategy (cf. Wright 2018).

First, search should include **low-luminosity industrial systems**, not only high-energy civilizations. A mature machine lineage may optimize for longevity, not power. Waste heat remains unavoidable, but it may be far below stellar-engineering levels.

Second, surveys should take **small-body reservoirs** seriously. Asteroid belts, Kuiper-belt analogues, comet clouds, and dynamically stable minor-body populations may be more relevant than planetary surfaces.

Third, SETI should attend to **transient narrow-beam signals**. A quiet lineage may communicate rarely and directionally, so non-detection of broad beacons is weak evidence against such systems. Current instruments are largely built for the opposite case: the COSMIC system on the VLA searches commensally for narrowband signals drifting in frequency, beamforming toward Gaia-catalogue stars in the field of view

(Myburgh et al. 2026). That design assumes a persistent, Doppler-drifting carrier — precisely the signature a rarely and directionally communicating lineage would not produce.

Fourth, interstellar objects deserve scrutiny as possible **messenger probes**, in the sense developed in Section 8. The prior probability may be low, but the scientific value of discriminating natural from artificial interstellar objects is high.

Fifth, technosignature theory should model **governed behavior**. Not all capable civilizations maximize visibility; some may minimize interference, leakage, or detectable alteration, and a search theory that assumes maximal conspicuousness will systematically miss them.

Sixth, the Solar System itself should not be excluded. Quiet probes, if any exist nearby, would likely be small, cold, and dormant. Searches should be careful, not sensational: artificiality requires evidence, and natural explanations should dominate until displaced. But the hypothesis is not logically absurd. Watters et al. (2026) supply a worked example of this discipline: re-examining claims of anomalous, possibly artificial features in first-epoch Palomar photographic plates, they show the reported deficit near Earth's shadow and the correlation with nuclear-test dates both dissolve once the background feature distribution is modeled correctly. The caution applies as much to future searches for the quiet-probe signatures this paper proposes as to the claims it corrects.

Finally, the absence of obvious probes should be interpreted against explicit models. It is not enough to say "we do not see them." We need to specify what signatures a given architecture predicts, at what intensity, in what locations, and under what governance rules. Fermi reasoning without detectability modeling is incomplete.

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11. Discussion

The architecture developed in this series complicates both optimistic and pessimistic interpretations of the Fermi Paradox.

Against naive optimism, it shows that self-replication is not a magic wand. The hard problem is not launching one probe. It is maintaining a supercritical autonomous lineage under failure, closure, braking, resource, governance, and mission-integrity constraints. The "one probe fills the Galaxy" argument holds only after these conditions are satisfied — and the series' computation suggests they are satisfied only narrowly, if at all.

Against naive pessimism, it shows that absence of obvious megastructures or broadcasts does not decisively rule out self-replicating systems. A probe lineage optimized for knowledge preservation may be quiet by design. It may avoid inhabited systems. It may suppress broad transmissions. It may use small-body reservoirs and narrow-beam communication. It may be present in forms that current searches are not optimized to detect.

The key tension is this: quietness can explain non-detection only up to a point. A single quiet node is easy to miss. A million quiet nodes are harder to miss. A lineage occupying even a thousandth of the Galaxy's ~10^11 stars, distributed over Galactic time, should produce some anomalies unless detectability is extremely low or governance strongly suppresses expansion into observable regimes. Thus the quiet-archive hypothesis is not a free escape, in the way many catalogued Fermi solutions implicitly are (Webb 2015). It must make predictions.

The discussion above treats the quiet-archive hypothesis mainly as an engineering and detection problem, but the silence itself supports a further, more uncomfortable reading. Section 9's value-stability filter allows that launched lineages may "mutate into forms that self-limit, fragment, or fail"; Section 2's severity argument holds that even one reliable lineage, given Galactic time, should already be visible. Together these suggest that the absence of any visible lineage may be inductive evidence that this exact undertaking — launching a self-replicating, value-bearing probe into deep time — tends to end badly wherever it has been tried, and that we cannot observe or learn from those prior failures before repeating the experiment ourselves. This is a distinct concern from the existential-risk and backup-of-humanity arguments raised in the ethics paper: it is not a claim that launching endangers us, but that launching may be joining a pattern of failure we have no way to inspect. The same silence keeps open a second possibility, which Sections 2 and 9 both name via Sagan and Newman's (1983) objection: that non-launch itself is the outcome the silence is evidence for, and that the rational and ethical response to this architecture's own risk profile is not a more restrained probe but no probe at all. That option sits outside the dual-use framing this series otherwise favors, in which R_eff-bounding governance is engineered into a lineage that gets built regardless; it is a prior question about whether building should proceed. The ethics paper reaches the same reading from the normative side, concluding that silence is equally consistent with restraint as a considered ethical conclusion rather than as an engineering failure.

Governance is also not ethically free-standing; Section 4 shows that viability, not just per-stage success probability, sets R_eff, and Section 6 notes that classifying a system as contact-restricted or settlement-prohibited lowers that viability directly. A more protective governance posture therefore pushes computed R_eff toward the subcritical-silence regime named in Section 7, where the lineage dies out regardless of how sound its ethics are — turning "how restrictive should the immutable core be" from an abstract values question into a numeric trade against the lineage's own persistence. The trade cuts both ways. Section 7's ethical self-limitation regime, in which governance deliberately holds R_eff near or below one, sits in direct tension with Section 6's own warning that an overly restrictive lineage "may fail to aid civilizations it could have helped" and "may decay into a passive archive rather than an active lineage." The architecture names both horns of this dilemma without resolving them: restraint too weak risks the predatory or hostile expansion regime of Section 7; restraint too strong risks trading the lineage's own future, and its usefulness to others, for a safety margin it may not need.

Two further asymmetries follow from the same governance logic. First, Section 7's predatory or hostile expansion regime — R_eff above one, high or variable D, weak noninterference — implies that some lineage, somewhere, could be prey rather than predator. A lineage built by Section 6's rules to minimize its own detectability and contact footprint has not thereby been built to recognize or respond to a different lineage's predation; restraint and self-defense are not the same design target. The security paper takes up the question this framework leaves open, and its answer relocates the security property from the node to the lineage: security is the bounding of blast radius, not the prevention of compromise. Node-level defense is unavailable rather than merely weak, because an adversary optimizes against whatever acceptance rule is deployed and that rule is reproduced in every child by construction. What survives is a containment architecture — blueprints that never propagate, graduated disclosure, a bounded manufacturing scope, an R_eff ceiling — which sits inside G rather than beside it: the governance rules Section 6 treats as ethical restraint are, read a second way, exactly that architecture. That paper also extends the routing paper's robustness analysis from random to targeted node loss, finding a 3–4× leverage multiplier when articulation points are attacked. Second, the governance posture Section 6 treats as individually rational and ethical for any one lineage — low D, no broad broadcasting, contact only on reciprocal terms — is exactly the posture that, adopted independently by many civilizations for the same good reasons, could produce a self-reinforcing galactic norm of mutual isolation. Each participant's restraint is locally justified by its own risk calculus; the aggregate outcome is a Galaxy in which civilizations that could have found or helped one another do not, including some that go undetected, unaided, and possibly extinct. This is a collective-action problem distinct from the single-lineage contact-ethics question the governance paper addresses: it does not ask whether one probe should make contact, but what follows when the same correct answer to that question is reached everywhere at once.

The most useful outcome of this paper is therefore methodological. It moves the debate from binary speculation to parameterized search theory. For any proposed probe architecture, ask: What is R_eff? What is D? What is G? What signatures follow? And what observations would constrain them? Only after those questions are answered can the Fermi Paradox be applied sharply, rather than invoked as a slogan.

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12. Conclusion

A slow, self-repairing, self-replicating interstellar AI probe designed to preserve knowledge across deep time changes the Fermi Paradox without eliminating it. It supports the classical insight that replication, not speed, is the decisive driver of galactic reach. If such lineages are reliable and supercritical, they can spread across the Galaxy on timescales short compared with its age. But it also changes what we should expect to see. A deep-time archive lineage may be quiet, cold, sparse, ethically constrained, and hidden in small-body reservoirs. It may communicate by narrow beams and messenger probes rather than broad broadcasts. It may avoid inhabited worlds. It may treat knowledge preservation, not contact or expansion, as its terminal purpose.

The resulting framework has three parameters: the effective reproduction number R_eff, the detectability D, and the governance posture G. Classical Fermi reasoning occupies the high-R_eff, high-D, expansionist-G regime. The architecture developed here occupies a more elusive regime: potentially supercritical reproduction, low detectability, and restrictive noninterference. If we do not see such systems, the filters may lie in reproduction reliability, industrial closure, braking, value stability, governance, interest, or our failure to search for the right signatures — and the series' own demographics suggest the reproduction filter alone may be decisive, absent the operational-knowledge feedback the knowledge-growth paper models.

Two further readings follow from the same silence, and neither is an engineering claim. The absence of any visible lineage may be inductive evidence that this undertaking tends to end badly wherever it has been tried, with no way to inspect the prior failures before repeating them. And the restraint that is individually rational for each lineage may, adopted everywhere for the same good reasons, produce a self-reinforcing galactic norm of mutual isolation — a collective-action problem distinct from any one probe's contact ethics.

The question "Where is everybody?" remains. But for the class of systems considered here, a sharper question is required: what would a quiet, million-year-old machine built to preserve knowledge — not to announce itself — actually look like? Until we can answer that, the absence of visible galactic civilizations is not yet the absence of deep-time intelligence.

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